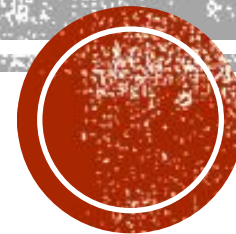


ASSIGNMENT 3



ASSIGNMENT

- Derive the use cases based on the requirements from CPS and draw use case diagram.



- Seven distinct tasks are carried out in requirements engg phase...
 - Inception
 - Elicitation
 - Elaboration
 - Negotiation
 - Specification
 - Validation
 - Requirements Management

- Under elaboration... we have use cases..



- During elaboration, the software engineer takes the information obtained during inception and elicitation and begins to expand and refine it
- Elaboration focuses on developing a refined technical model of software functions, features, and constraints
- It is an **analysis modeling task**
 - Use cases are developed
 - Domain classes are identified along with their attributes and relationships
 - State machine diagrams are used to capture the life on an object
- The end result is an analysis model that defines the **functional, informational, and behavioral** domains of the problem



WHAT IS USE CASE?

- Primary actors and use cases are identified, and key scenarios (those that will be implemented first) are detailed.
- Each use case has an associated use-case specification.
- Identifies actors involved in an interaction and names the type of interaction
- This is then supplemented by additional information describing the interaction with the system
- The set of use cases represents all of the possible interactions that will be described in the system requirements



USE CASE..

- Step One – Define the set of actors that will be involved in the story
 - Actors are people, devices, or other systems that use the system or product within the context of the function and behavior that is to be described
 - Actors are anything that communicate with the system or product and that are external to the system itself
- Step Two – Develop use cases, where each one answers a set of questions



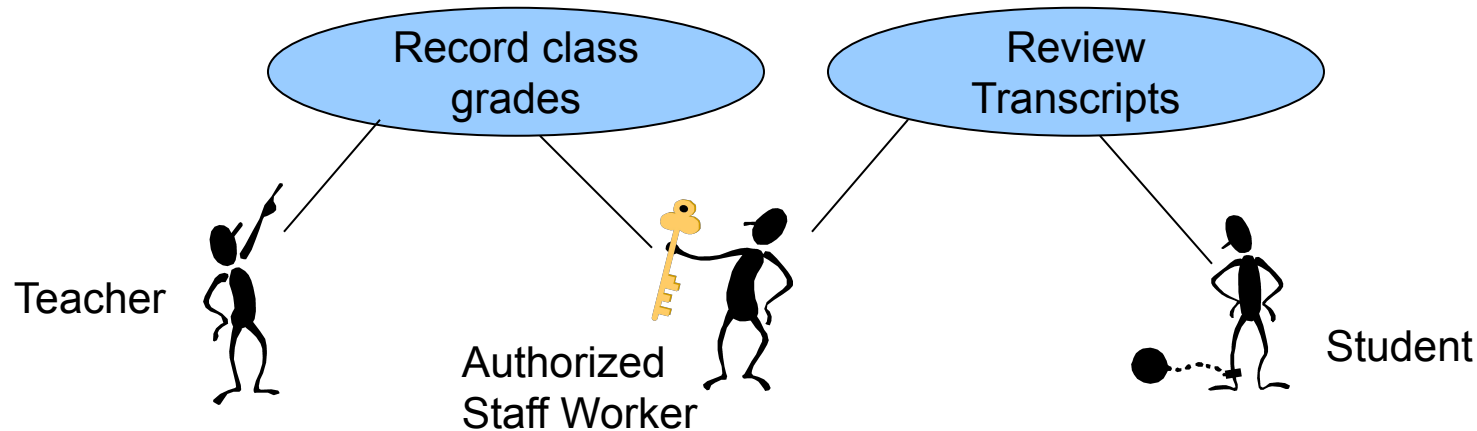
QUESTIONS..

- Who is the actor(s)?
- What are the actor's goals?
- What preconditions should exist before the scenario begins?
- What main tasks or functions are performed by the actor?
- What exceptions might be considered as the scenario is described?
- What variations in the actor's interaction are possible?
- What system information will the actor acquire, produce, or change?
- Will the actor have to inform the system about changes in the external environment?
- What information does the actor desire from the system?
- Does the actor wish to be informed about unexpected changes?



WHAT IS USE CASE DIAGRAM..??

- A diagram that shows set of use cases and actors and their relationships.
- They represent systems functionalities..
- The requirements of the system from user's perspective.



NOTATIONS

- Use case
- Actors
- Include
- Extends



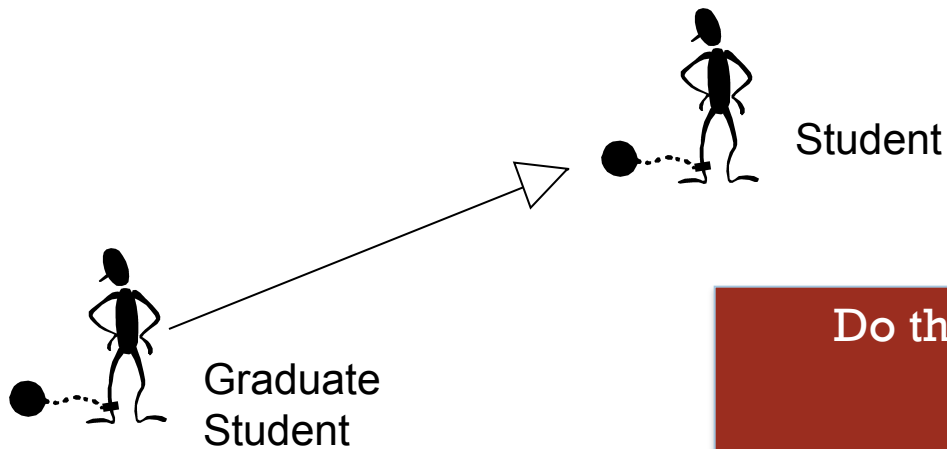
- **Actors:**
- Actors are people or external systems that need to interact with our system

Finding Actors

- Who or what will use the main functionality of the system?
- Who or what will provide input to this system?
- Who or what will use output from this system?
- Who will need support from the system to do their work?
- Are there any other software systems with which this one needs to interact
- Are there any hardware devices used or controlled by this system?



- Actors can be related by generalization/specialization
- Actors are classifiers (not individual users) do not say – Rohan, Riya..
- Do not show this level of communication between actors.
- Name them singular.



Do this when very obvious..
otherwise skip it



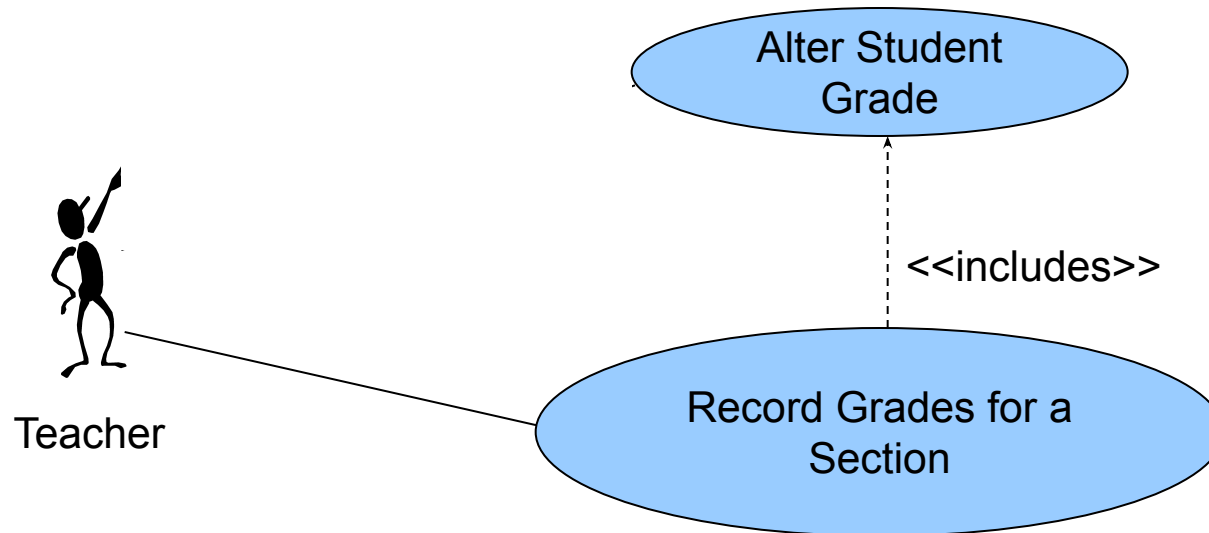
- Use case:
- A description of set of actions including variants
- Use case should begin with a verb, eg : generate report.
- Avoid keeping it open ended. i.e say register new user instead of just register.





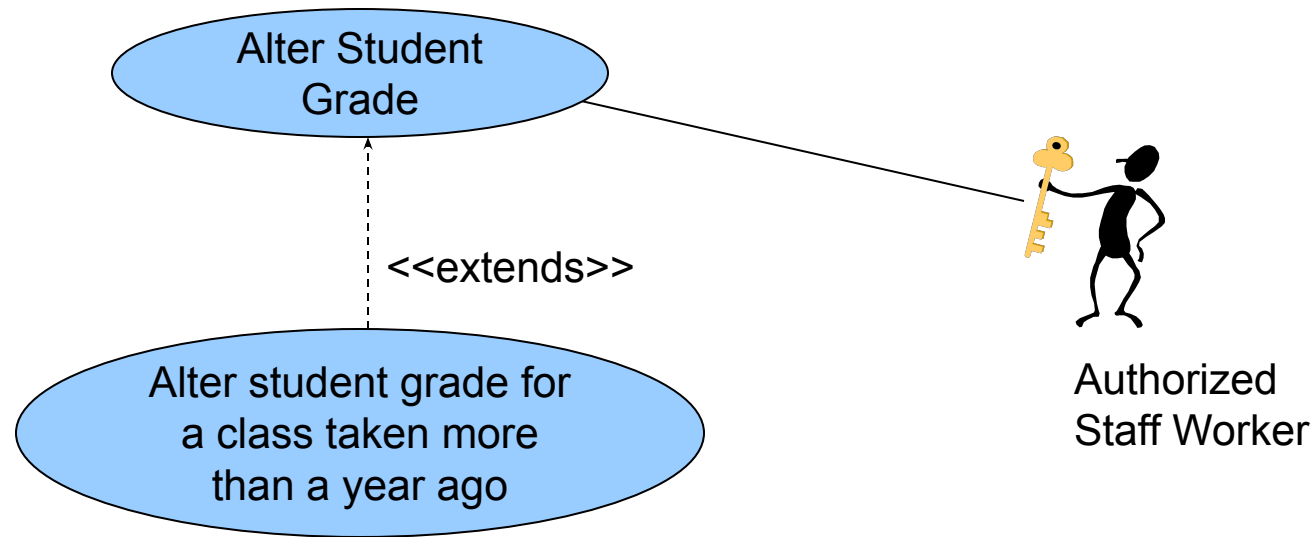
INCLUDES..

- *Includes Dependency*: Defines how one use case can invoke behavior defined by another use case

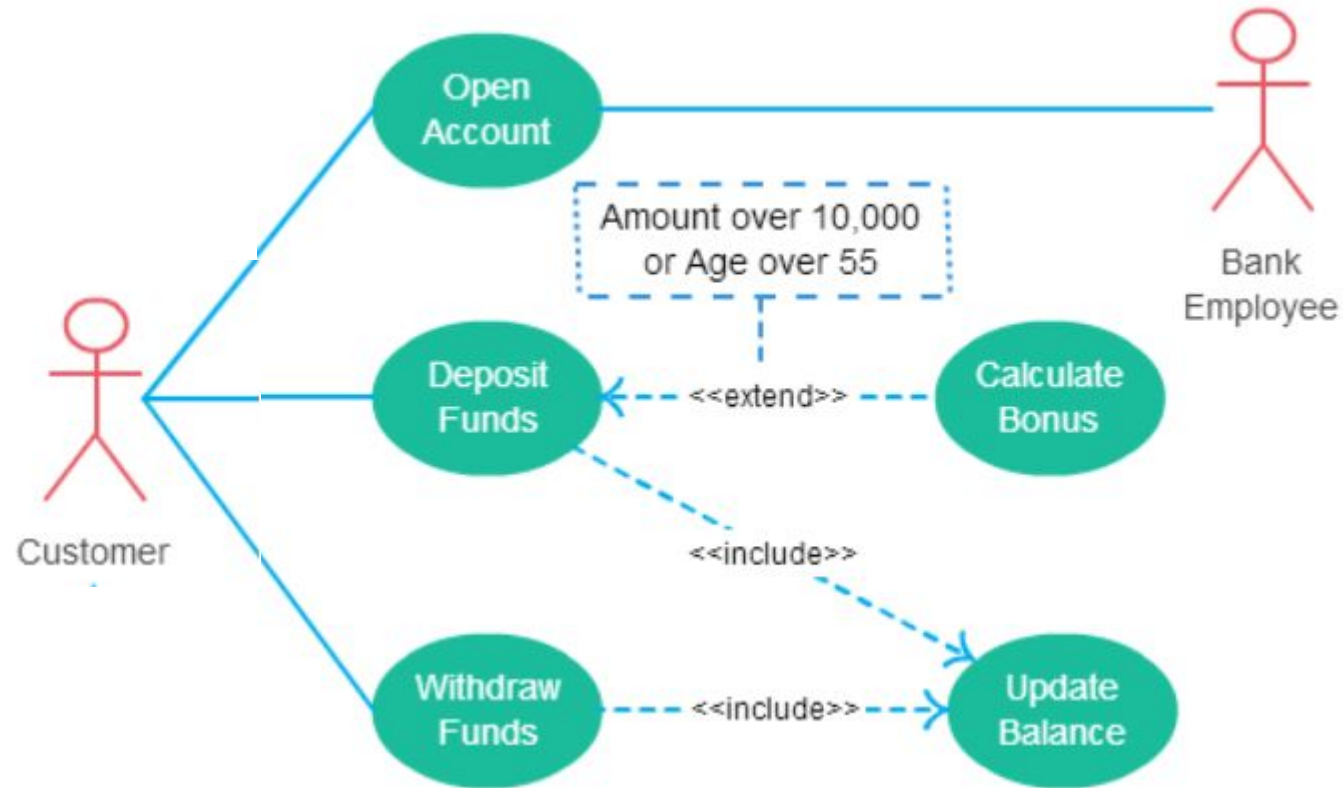


EXTENDS

- *Extends dependency*: defines a use-case that is a variation of another, usually for handling an abnormal situation or can be taken up as additional functionality



INCLUDE/EXTENDS.. ONE MORE CASE:

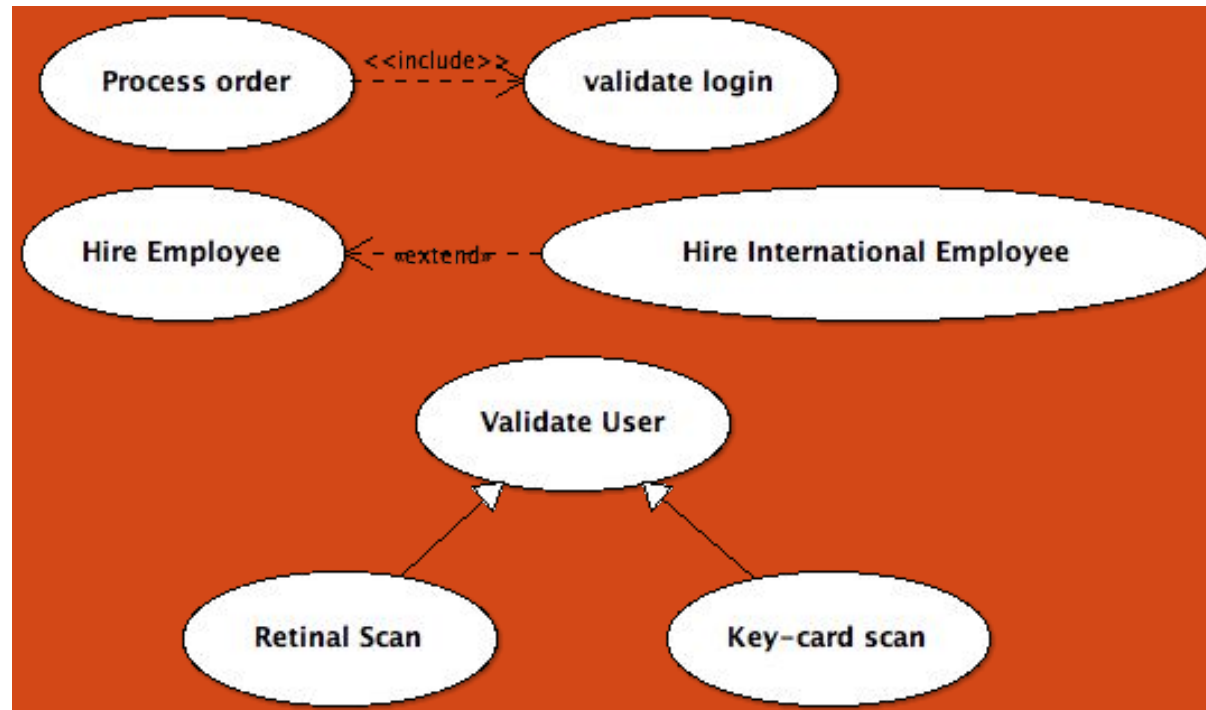


INCLUDE/EXTEND/GENERALI ZE

Includes

Extends

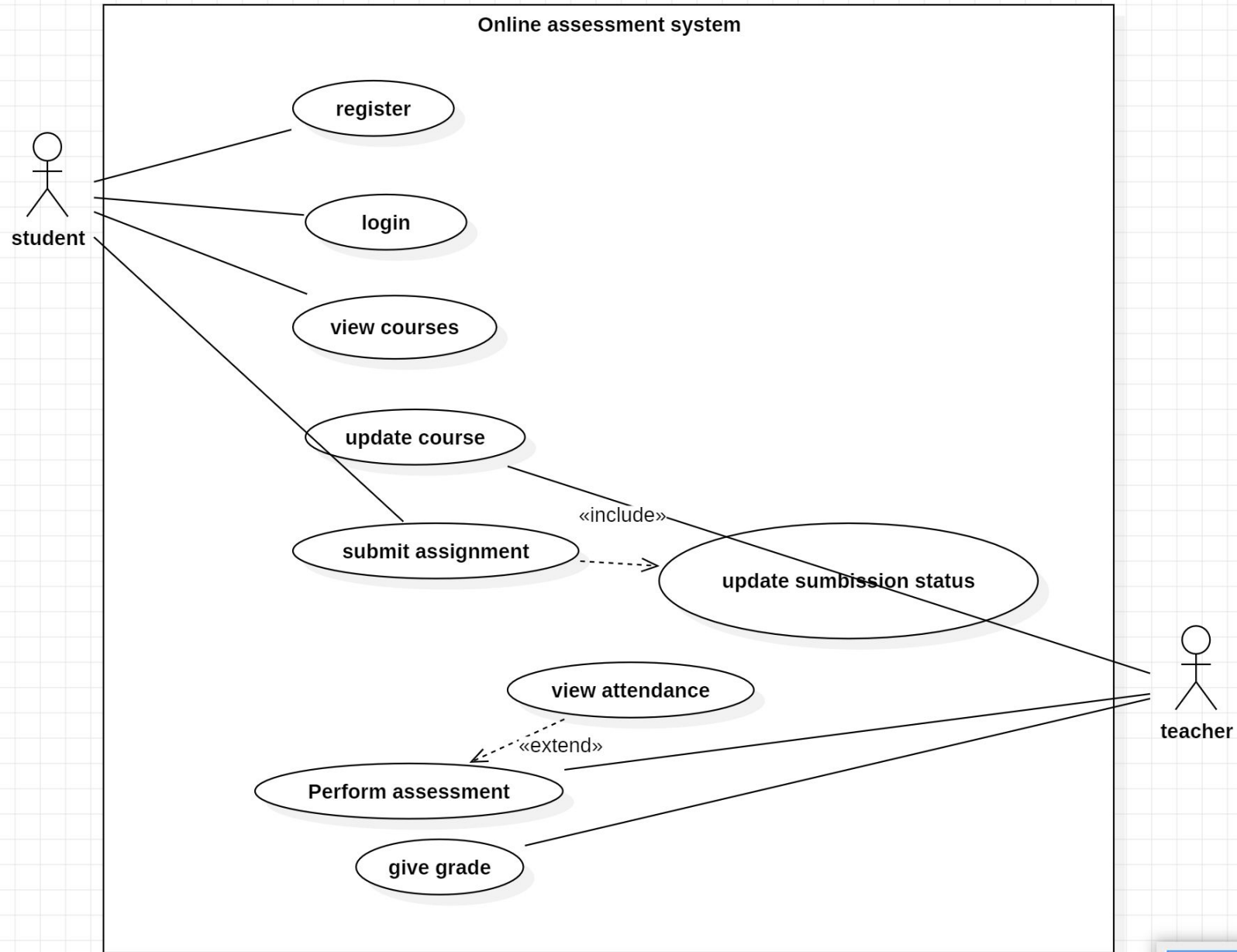
Generalization



STARUML

- Install Evaluation version ...
- Example : online assessment system..





WHAT IS IT THAT WE HAVE TO DO??

- Any product/ system.. Based on CPS..
- For the function requirements generate use case diagram..
- Questions??
- Whether it should be fine or course grained??
- It should embody sufficient levels of granularity without which it would be rendered useless.

